Space Weather Events – Teacher Copy

Authors: Kira Lee, University of Tasmania

In this class, students analyse magnetograms and Kp index data from three different geomagnetic events to identify patterns and variations in geomagnetic activity.

They then focus on the St Patrick's Day 2015 geomagnetic storm as a case study, using coronagraph images to estimate the speed and arrival time of the associated coronal mass ejection (CME), then compare these to the actual values.

Australian Curriculum: Science F-10 Version 9.0 links:

Use equipment to observe, measure and record data with reasonable precision, using digital tools as appropriate (AC9S5I03 AC9S6I03).

Analyse and connect a variety of data and information to identify and explain patterns, trends, relationships and anomalies (AC9S9I05 AC9S10I05).

Pre-requisite knowledge:

- Students should have a conceptual understanding of the Earth's magnetic field and geomagnetic activity.
- Students should be familiar with time series data and be able to identify trends and peaks.
- Students should be familiar with the v = s/t equation and be able to convert between units of time and distance e.g., mm to km with guidance.

Aims:

- Identify space weather events using magnetograms and Kp data
- Estimate CME speed and arrival time from a series of coronagraph images
- Compare their estimations to the actual St Patrick's Day 2015 CME event and consider sources of error in their calculations

Materials:

- Paper copy of the document CME_Image_Set.docx (ideally print this document in colour and ensure it is printed to its original scale)
- Ruler
- Pencil
- Calculator

A smart device with access to Google Sheets/ Excel is also useful for the CME calculations (but not essential).

Space weather events are events driven by distant solar activity that impact human technology and the near-Earth space environment (BoM SWS). In this class, you will use real scientific data and plots to identify space weather events and then study one of these as a case study to calculate the speed and arrival time of the associated coronal mass ejection.

Activity 1: Identifying space weather events using magnetograms & Kp data

You have been provided with 3 sets of plots in the document: **Space_Weather_Events_Plots.pdf**.

These correspond to 3 different 2-week time-periods:

A: 03-16 May 2024

B: 10-23 March 2015

C: 01-14 November 2019

The top plots for each time-period are **magnetograms**. Magnetograms are time-series plots that show variations in Earth's magnetic field in three orthogonal directions – x (north-south), y (east-west) and z (up-down) (BoM SWS). Measurements are commonly reported in nanotesla (nT) after Serbian-American inventor Nikola Tesla (Ungvarsky, 2023).

Geomagnetic storms are seen as **spikes** in magnetogram plots i.e. the lines suddenly fall/ rise to a minimum/ maximum value and then return to relatively flat (quiet) conditions.

The **x component** of magnetograms refers to geomagnetic disturbances in the magnetic north direction. This component is most relevant for identifying geomagnetic activity as geomagnetic storms cause temporary shifts in the position of magnetic north (NASA, 2015).

The bottom plots are time series plots of **Kp index** value. The Kp index is a measure of global geomagnetic disturbance on a scale from 0-9 taken at 3-hour intervals (SWPC NOAA). According to the NOAA space weather scale, a minimum Kp value of **5** is needed to classify an event as a geomagnetic storm.

For each event answer the following questions:

1. Was there a geomagnetic storm?
2. How do you know if there was a storm (or not)?
3. What date was peak geomagnetic disturbance?
4. Quantify peak geomagnetic disturbance (x component)/ nt
5. Max Kp index value reached
6. NOAA G-scale classification
7. What impacts do you think this event had?

These questions are listed in **Table 1** below.

You will need to refer to the NOAA space weather scales (available here or in the document **NOAAscales.pdf**) to answer the questions. This document contains information on the predicted impacts of space weather events at each level on the scale.

Try not to reveal to students that the 2015 event (studied in Activities 2 and 3) is a geomagnetic storm event before they have completed Activity 1, as this will reveal the answers to some of the questions in the table.

The impacts of events A and B listed in **Table 1** are taken from published literature about the specific geomagnetic storms. The impacts students predict using the NOAA space weather scale will differ/ not be as specific. Encourage them to look at the cited sources when they go through the answers for this class.

Table 1: Interpretations of the geomagnetic conditions for events A, B and C ANSWERS

	A 03-16 May 2024	**B** 10-23 March 2015	**C** 01-14 November 2019
Was there a geomagnetic storm?	YES	YES	NO
How do you know if there was a storm (or not)?	Large disturbance seen in the magnetograms around 11 May Kp rises to a value of 9	Large disturbance seen in the magnetograms around 17/18 March Kp rises to over 7	No big spikes in magnetogram data Kp < 5 throughout the time-period
What date was peak geomagnetic disturbance?	11 May 2024	17 May 2015	n/a
Quantify peak geomagnetic disturbance (x component)/ nt HINT: calculate the max-min x value	23100 – 22600 = **500 nt**	23250 – 22900 = **350 nt**	23160 – 23100 = **60 nt** (This disturbance is not due to geomagnetic storm activity; it is likely just normal daily variation in Earth's magnetic field)
Max Kp index value reached	9	7.6	2.6 (Not storm level)
NOAA G-scale classification	G5	G3/4	n/a
What impacts do you think this event had?	GICs (Ground Induced Currents) Satellite drag & satellite charging GPS & GNSS errors Degradation of HF communications Auroras at unusually low latitudes (visible as far north as Townsville, 19.2599° S!) Plane diversions (Horne et al., 2024) (Bowler, 2025)	GICs (Ground Induced Currents) Low latitude auroras Strong GNSS/ GPS disturbances Disrupted satellite and radio communications (NOAA SWPC, 2015) (Jacobsen and Andalsvik, 2016)	n/a

We are now going to consider the March 2015 event as a case study.

Activity 2: Calculating the speed & arrival time of the St Patrick's Day 2015 CME

Event B (March 2015) was a geomagnetic storm event named the **St. Patrick's Day solar storm**. We will be analysing the motion of the CME associated with the St Patrick's Day storm using a series of coronagraph images from NASA's Solar and Heliospheric Observatory (**SOHO**). The images are found in the file **CME_Image_Set.docx**.

Coronagraphs are specialised space-based instruments designed to block out the Sun's light so the corona (the Sun's outermost layer) can be viewed (Mann, 2019). Images from coronagraphs are used to monitor space weather events like coronal mass ejections (CMEs).

The white circle on each image is the outline of the Sun. The solid red disk is a suspended plate that blocks out the most intense sunlight, allowing SOHO to perceive the CMEs. A date and time stamp (in UT) can be found in the bottom left-hand corner.

The SOHO video recording of this CME can be found here.

1. On each image, mark a feature of the CME that you can trace between images. The shape of the CME changes over time like a cloud, so try to mark a feature at the front of the CME.

See the document **CME_Image_Set_TEACHER.docx** where the edge of the CME cloud has been marked in each image. This isn't an exact science. Different features to the one marked can be used to track the CME movement.

2. With a ruler, measure the distance (in mm) between the CME feature that you are tracking and the edge of the Sun for each image (see **Figure 1** as an example).

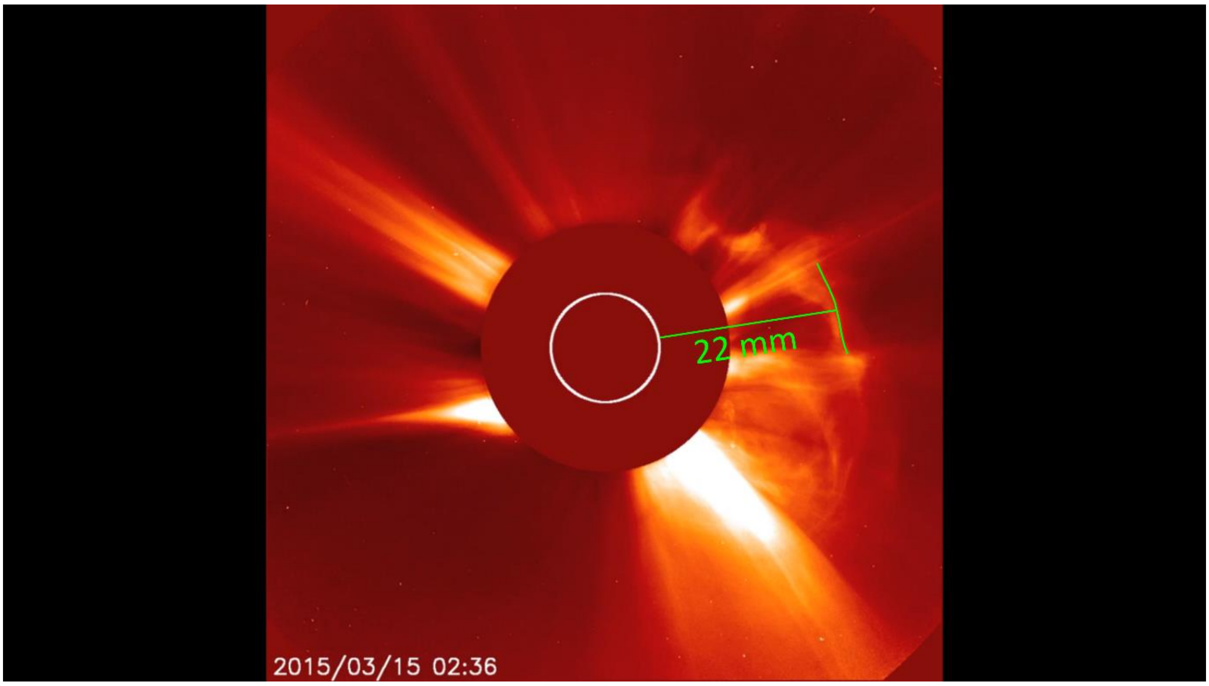

Figure 1: Example CME measurement (not to scale). Image courtesy of SOHO/LASCO consortium. SOHO is a project of international cooperation between ESA and NASA.

3. Add these positions to **Table 2** below and fill out the time columns.

Table 2: St Patrick's Day 2015 CME position, time and velocity calculated from the coronagraph images EXAMPLE ANSWERS

Image Number	Position (mm)	Clock time	Time between images (hrs)	Total elapsed time (hrs)	Average velocity (mm/hr)
1	0	01:36	0	0	0
2	10	02:00	0.4	0.4	25
3	15	02:12	0.2	0.6	25
4	22	02:36	0.4	1	22
5	34	03:12	0.6	1.6	21.25

These calculations and answers are also provided in the Excel spreadsheet **CME_calculations.xlsx**.

Students will have different position (mm) measurements because they will be following different features on the coronagraph images. The time columns should all be the same.

4. Calculate the CME velocity for each image using v = s/t, where s is the position (mm) and t is the total elapsed time (hrs). Add these values to **Table 2**.

Students should divide the value in the position column by the value in the total elapsed time column. They will all have different values, but the general trend should be that the CME decreases in velocity in time.

The velocity of the CME varies through time as the CME cloud expands as it moves towards Earth. We will take the velocity you calculated for **Image Number 5** as the velocity of the CME as it travels towards Earth. Make a note of this velocity (including units).

While using mm/hr is a convenient way to study general motion, it is more meaningful to convert these to km/hr in order to estimate how long it will take the CME to reach Earth.

5. Measure the diameter of the Sun in your images in mm. The actual diameter of the Sun is **1,391,400 km**. How many km is represented by one mm? This is your scale.

In the images, the diameter of the Sun measures approximately 14 mm.

1,391,400 km/ 14 mm = 99385.7 km

1 mm on the images represents approximately **99385.7 km**

6. Convert your velocity for Image Number 5 from mm/hr to km/hr using the scale you created from measuring the Sun.

The velocity calculated from Image Number 5 is 21.25 mm/ hr.

21.25 mm/hr * 99385.7 km = **2111946.4 km/ hr** (the exact value will vary between students)

The general velocity range for CMEs is 900,000 km/hr to 10,800,000 km/hr so anything between this is acceptable.

The standard unit of CME speed is **km/s**.

7. Convert this velocity from km/hr to km/s.

HINT: there are 3600 seconds in an hour.

2111946.4 km/hr / 3600 s = **586.65 km/ s**

Now that we have a value for the speed of the St Patrick's Day 2015 CME, let's calculate what time it should have reached Earth's magnetic field.

8. The Sun is an average of **149,600,000 km** from Earth. Using the speed (km/ hr) you calculated for Image 5, how many hours will it take this CME to reach Earth? Assume a constant speed.

HINT: t = s/v

149,600,000 km / 2111946.4 km/ hr = **70.8 hours** (the exact value will vary between students)

9. This CME erupted from the Sun at around **02:10 15/03/2015** UT (Wu et al., 2016). Using the number of hours you calculated above, estimate the date and time that this CME should have arrived at Earth. Feel free to use online tools to help you.

The CME will take approximately 70.8 hours (approximately **3 days**) to reach Earth.

This gives an arrival date/ time around **00:58 on 18/03/2015**.

Activity 3: Comparing your estimations to the actual St Patrick's Day 2015 event

We will now compare your estimated CME speed and arrival time from **Activity 2** to the actual values for the St Patrick's Day 2015 geomagnetic event.

The actual initial propagation speed of the St Patrick's Day CME was **668 km/s** (Wu et al., 2016).

1. How close was your estimated speed to the actual value?

The CME images give a speed around **586.65 km/ s**

Any value between 500-700 km/s is a good estimation.

This CME hit Earth's magnetic field around **04:45 17/03/2015** (Murphy et al., 2018).

2. How close was your estimated CME arrival date/ time to this?

The CME images give an arrival time of around **3 days** (70.8 hours), which gives a storm arrival time around **00:58 on 18/03/2015**.

Any time within 24 hours of 04:45 17/03/2015 is a good estimation.

3. Can you think of any assumptions we made or any sources of errors in our calculations that would explain why our estimated values differ from the actual?

- We assumed that the CME maintained a constant speed after coronagraph image 5 (which is not the case, CME speed would have continued to vary through time).

- We used an approximate diameter of the Sun as our distance scale, which will not have been perfectly accurate.

- We assumed a constant distance between the Earth and the Sun, which is a source of error as, in reality, this distance constantly.

- We only considered 5 images from the first few hours of this CME's journey towards Earth. To get more accurate speed and arrival time values, we would need to consider this CME at later propagation stages.

- The feature you identified to track the CME changed over time and is not a perfectly reliable indicator of the motion of this CME.

- There may have been human errors in our measurements.

The interesting thing about this storm is that it arrived **15 hours ahead** of forecast and was only predicted to be a **G1** level storm (NOAA, 2015)! This shows how important accurate geomagnetic forecasting is.

Additional information about the 3 geomagnetic events you considered in this class can be found in the document **Storm_Event_Information.pdf**.

You can view current geomagnetic conditions and forecasts from the Australian Space Weather Forecasting Centre here and the NOAA here.

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